

Whittaker Vanishing And Archimedean Repair

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Summary

Pass 82 tests the residue of the maximally degenerate principal series

$$I(s) = \chi_s$$

under the unipotent radical $U = \epsilon$. Because $I(s)$ is trivial on U , there are no nontrivial Whittaker functionals:

$$\mathrm{Hom}_U(I(s), \psi) \cong \begin{cases} R, & \psi = 1, \\ 0, & \psi \neq 1. \end{cases}$$

Only the constant term survives.

Main Claims

- **Whittaker vanishing.** Nontrivial finite additive characters exist at every level, but their coefficients against the constant U_N -action vanish. They do not survive as a solid Whittaker model for $I(s)$.
- **Rosser carrier.** The Rosser torsor is not a generic Whittaker coefficient. It is the unipotent shear parameter $U = \epsilon$ itself.
- **Archimedean repair.** Adding the real place gives the adelic solenoid

$$\Sigma = (\mathbb{R} \times \widehat{\mathbb{Z}})/\mathbb{Z} \cong \mathbb{A}/\mathbb{Q}$$

and, after the Pass 83 correction, the compact exact sequence

$$0 \rightarrow \widehat{\mathbb{Z}} \rightarrow \Sigma \rightarrow \mathbb{R}/\mathbb{Z} \rightarrow 0.$$

The finite phantom $\epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}$ is the dense quotient $\Sigma/\mathbb{R}$, not the closed kernel of the projection to $\mathbb{R}/\mathbb{Z}$. This repairs global adelic Fourier duality, but it does not create a finite-prime solid morphism $\epsilon \rightarrow \mathbb{Q}$.

Machine Verification

`code/scripts/check-pass82.py` produced `artifacts/reports/pass82-whittaker-archimedean-repair-check.json` with overall PASS. It checks vanishing of nontrivial finite Fourier coefficients, the U -equivariant Hom dimension table, and finite shadows of the solenoid exact sequence.

Next Use

Pass 83 should compare the global solenoid Σ with the finite phantom ϵ as an exact triangle and decide whether global Fourier theory restricts to only a constant term or also yields a boundary class measuring the lost finite-prime Weyl flip.